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Chemostratigraphic Delineation of Sedimentary and Igneous Lithologies During Drilling without MWD Measurements

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Abstract

The identification of igneous basement rocks from ditch-cuttings samples can be challenging, due to cuttings sizes and textures generated by PDC-bits and turbine drilling. Thus, basement rocks are sometimes mistakenly categorized as fine-grained sedimentary lithologies. With the absence of measurement while drilling (MWD) logs, basement identification becomes very challenging. Advanced mudlogging data, i.e. X-ray fluorescence (XRF; elements) and X-ray diffraction (XRD; mineralogy), are capable of overcoming this challenge when understood properly.

Geochemical data (XRF) are used to model sedimentary lithologies (element ratios, proxies, and simulated gamma ray logs). The data are further linked to underlying mineralogy (XRD and element:mineral proxies). Geochemical indices and provenance discrimination function analysis logs and diagrams indicate different rock types, e.g. sedimentary versus igneous. Both, geochemistry and mineralogy are then matched to standard igneous rock compositions. Additional methods, if available, such as petrography of cuttings, can be integrated and increase confidence in the chemostratigraphic interpretation. Post-drilling analyses, e.g., wireline logs, image logs, and sidewall cores will verify the results or refine the model.

Standard mudlogging cuttings geological descriptions from a well drilled without MWD support were ambiguous and incapable of identifying basement rocks while drilling. Well-site generated XRF and XRD data interpretations (chemostratigraphy) indicated different sedimentary lithologies, as well as penetration of mafic igneous rocks. Petrography on cuttings thin-sections agrees with the geochemical/mineralogical findings. Subsequently, a bottom core (diorite) was cut and the well was terminated. Post-drilling petrography and lab-based XRD on sidewall cores, as well as wireline logs confirmed the modelled lithology types and interpreted basement rocks from the cuttings samples. This geochemical approach has successfully been applied to several wells (with and without MWD) encountering both, mafic and felsic igneous basement rocks. Challenges, however, are encountered with the presence of a weathering zone above the basement complex or sedimentary rocks derived from an adjacent basement complex, which show very similar geochemical signatures. Nonetheless, scanning electron microscopy petrography on cuttings verified the initial chemostratigraphic interpretation.

The identification of basement rocks while drilling without (risk minimalization or cost optimization) and even with MWD measurements is often challenging, if not impossible. Interpretation of advanced

mudlogging XRF/XRD data (low risk, cost-efficient) lowers the uncertainties in recognizing the penetration of basement rocks that agrees well with post-drilling analyses and thus assists in determining the well's terminal depth (TD).

Introduction

In some cases, Cost optimization or drilling hazards force drilling exploration wells without Measuring/Logging While Drilling (MWD/LWD) tools as part of the Bottom Hole Assembly (BHA). Rock sections, therefore, only get logged after drilling, leaving an evaluation of the strata and well placement interpretation while drilling to the ditch cuttings descriptions of a trained mudlogger and interpretations of a wellsite geologist.

Such cuttings descriptions may be hampered by the processes involved in the cuttings formation, namely Polycrystalline Diamond Compact (PDC) drill-bits and "mudmotor" drilling, both affecting the textures of the rocks to become unrecognizable.

The description of cuttings samples can therefore become indistinct and not reflect the original rock textures leading to misinterpretation of the drilled strata. For instance, it becomes challenging to differentiate fine-grained sand-/siltstones from finely crushed coarser sandstones, conglomerates, or crystalline igneous rocks, when ground to a fine-grained mix or even powder.

A cost-effective method, which also becomes standard for some mudlogging services, is the geochemical and mineralogical determination of rock cuttings compositions with X-ray fluorescence (XRF) and X-ray diffraction (XRD) analyzers. Because the geochemical and mineralogical composition is not significantly affected by the drilling process, these data provide valuable additional information from the drilled strata in near-real time.

Portable XRF and XRD analyzers at the wellsite give a quick overview of the geochemical and mineralogical composition of cuttings samples and are used for chemostratigraphy for characterization of the strata and well placement (e.g., [El Gezeery and Scheibe, 2010](#), [Marsala et al., 2011](#)).

However, relatively little effort has been devoted to the differentiation of igneous basement from sedimentary rocks, because the majority of exploration wells have not penetrated into basement rocks.

Nonetheless, a number of legacy wells drilled without MWD show non-desired deeper penetration of igneous basement rocks, based on misleading visual descriptions of altered cuttings samples, due to aforesaid drilling processes. Post-drilling analysis with XRF and XRD of cuttings from such wells revealed that it is possible to identify different types of igneous rocks.

The well presented in this study demonstrates the capability of wellsite XRF and XRD to overcome and identify igneous rocks while drilling. The interpretations are supported by in parallel performed petrographic thin-section analysis of a number of cuttings. Post-drilling wireline logging, as well as lab-based XRD and thin-section analysis of selected sidewall cores confirm the initial interpretation based on cuttings.

Statement of Theory and Definitions

Theory and rationale

Ditch cuttings "are the small pieces of rock that are chipped away by the bit while a well is being drilled. The rock fragments are transported by the mudstream from the bit to the surface where they can be "caught" and analyzed" (AAPG Wiki, 2022).

Such cuttings, however, are often visually indistinguishable when drilled with PDC bits and or the use of a motor. These processes tend to produce cutting that are either too small or too mashed up for identification of textures or the original rocks framework. Therefore, it becomes challenging to differentiate fine-grained sand-/siltstones from ground coarser sandstones, conglomerates, or crystalline igneous (basement) rocks, when ground to a fine-grained mix or even powder.

Basement rocks are defined in this study as igneous and/or highly metamorphosed igneous (ortho-metamorphic) rocks, that lie above the earth mantle and beneath all other rocks and sediments, also referred to as crystalline basement.

Such PDC and/or motor drilled cuttings can thus be misinterpreted during visual inspection affecting the well positioning assessment.

The cuttings, however, preserve their mineralogical and thus geochemical compositions.

Portable XRF and XRD analyzers are frequently used at the wellsite, as part of an enhanced mudlogging service, measuring the geochemical and mineralogical compositions of the drilled rocks.

XRF analyzers determine the inorganic geochemical composition of the cuttings/rock samples and provide a data set of c. 30 to 50 chemical elements depending on lithologies and instrument calibrations. The data encompass the ten major elements (Si, Ti, Al, Fe, Mn, Mg, Ca, Na, K, and P) and sulfur (S), commonly reported in their oxide form as weight percent (%), as well as a number of trace elements reported in parts per million (PPM). Major elements are the general the building blocks of the minerals, while the trace elements commonly substitute for major elements and are common in refractory minerals.

Because major elements can occur in a number of minerals (e.g., K may be hosted in K-feldspars, illite, micas, and others), they are not diagnostic for a specific mineral. Systematic variations, however, can give an indication of major changes in the mineral assemblages.

Wellsite XRD analyzers are commonly setup for an automated mineral assemblage output, and the technical staff at wellsite may not have the expertise for in-depth individual mineral analysis. Thus, wellsite XRD mineralogy data often comprise of the expected standard mineral assemblages commonly found in siliciclastic sedimentary rocks (e.g., calcite, dolomite, siderite, anhydrite/gypsum, quartz, albite (representing all plagioclase varieties), K-feldspar (all alkali-feldspars varieties), kaolinite, illite/mica, chlorite, smectite, halite, rutile/anatase, pyrite, and hematite).

More uncommon minerals, such pyroxenes, amphiboles, and olivine, are not part of these standard lists, because they are usually not preserved in deeply buried siliciclastic sediments. They are, however, important for identifying an igneous rock provenance. Nonetheless, systematic variations, can give an indication of major changes in the mineral provenance.

Un-weathered or unaltered (fresh) magmatic rocks have their initial mineralogical and geochemical compositions preserved, which differs from (weathered/altered) sedimentary rocks, although the latter may have been derived from these.

Igneous rocks can broadly be classified based on their mineralogical and hence geochemical compositions. A common classification is based on the silica concentrations (SiO_2) that differentiates acidic, intermediate, basic, and ultrabasic compositions with decreasing silica content. In addition, in magmatic rocks K_2O and Na_2O are mainly linked to their feldspar content and type (i.e., alkali- and plagioclase feldspars), which change from acidic to mafic/ultramafic rocks. FeO and MgO are associated with mafic minerals, with their amount increasing from acidic to mafic/ultramafic igneous rocks. The mineralogical variations and associated geochemical expressions are summarized in [Fig. 1](#) (redrawn after Earth Science Australia/earthscience.org).

Feldspars have different resistances towards weathering and diagenetic alterations. The plagioclase series shows increasing resistance from Ca-rich (e.g. anorthite) to Na-rich (e.g. albite) varieties. The alkali-feldspars (K-rich feldspars, e.g. orthoclase, sanidine, microcline) are generally relatively stable during these processes. Elements from dissolved feldspars, however, also precipitate as clay-minerals, e.g. the illite/mica and smectite groups.

These mineral properties and behaviors during alteration processes may provide indirect proxies and geochemical signals that can potentially be used to differentiate igneous from sedimentary rocks.

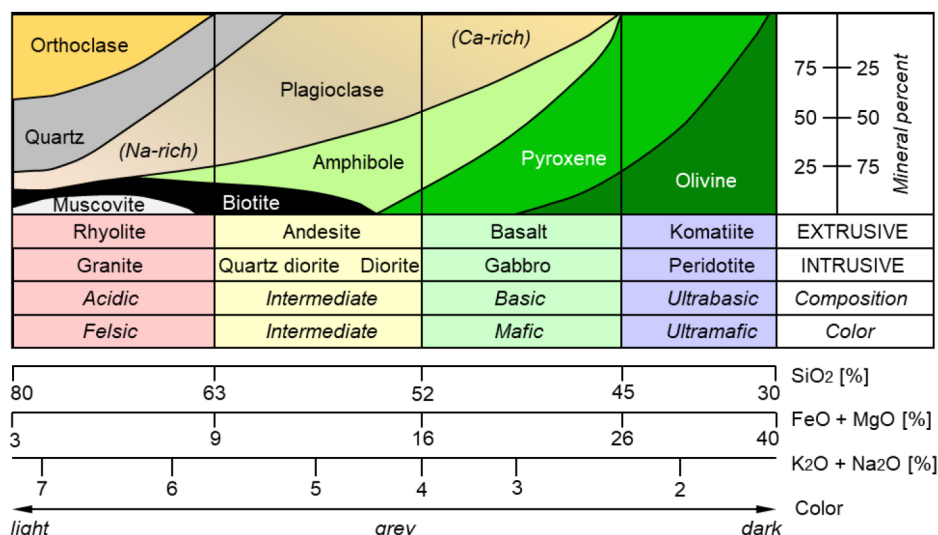


Figure 1—A geochemical classification of igneous rocks based on their general mineral assemblages and associated geochemical compositions (after Earth Science Australia/earthscience.org).

Workflow

The study well drilled into igneous basement rocks. The well is used as an example of the workflow to differentiate sedimentary from igneous rocks in the absence of petrophysical borehole measurements while drilling (MWD) for evaluation while drilling.

Geological descriptions of cuttings samples at the wellsite were hampered, due to the format of the cuttings.

The geochemical and mineralogical compositions, as determined at the wellsite, are utilized to delineate sedimentary from magmatic/basement rocks. In addition, geochemical indices, such as CIA, PIA, and WIP, and DF discrimination diagrams have been utilized in support (see below).

Generating thin sections for petrographic evaluation is usually not performed at wellsite, as the process is time-consuming and would require a petrographer at the wellsite. Petrographic thin sections have, however, been generated at the company's laboratory from cuttings, well bore bottom core, and (post-drilling) sidewall core (SWC) samples.

Description and Application of Equipment and Processes

Cuttings samples have been collected at surface by a mudlogging service company. Based on drilling and bore hole parameters, the depth for which the cuttings are received at surface is calculated, and samples are collected in intervals of 10 ft. Thus, a cuttings sample comprises of the mixed material from 10 ft. drilled borehole.

The material is cleaned, sieved and screened any foreign material, such as metal fragments from the BHA or 'caved' in rocks from previous drilled sections.

The material is then dried, and ground into a homogenous fine powder for XRF and XRD analyses.

All XRF data have been generated at the wellsite by a service company providing advanced mudlogging services, which include samples description under a binocular microscope, as well XRF and XRD analyses. The service company utilized a benchtop Energy-Dispersive XRF (ED-XRF) analyzer and reported 32 chemical elements.

For XRF analyses the powder is pressed into 32 mm wide and > 2 mm thick XRF pellets. A Spectro XEPOS XRF analyzer determined the chemical composition for under vacuum environment; reported 31 elements above detection limits in the range from Na to U of the PSE.

The XRD data from cuttings have been produced with a benchtop XRD goniometer at the wellsite. Eleven different minerals have been reported. The mineral list, however, consisted of a standard assemblage commonly suitable for siliciclastic and carbonate rocks. The utilized well site XRF instrument was not calibrated to differentiate alkali- from plagioclase feldspars (e.g., orthoclase (K), albite (Na), or anorthite (Ca)). Less common minerals in rather mature sediments, but typical for igneous rocks were omitted from analysis.

XRD data from post-drilling SWC's were prepared and analyzed at the in-house core laboratories.

Petrographic thin-section from cuttings and sidewall cores have been prepared at the inhouse laboratories and described by a professional petrographer utilizing a petrography polarization microscope.

Presentation of Data and Results

The study well 'Well A' was drilled without petrophysical logging tools while drilling (LWD/MWD).

While drilling, a service company provided mudlogging services, which besides others, included geological sample description, and XRF and XRD analyses. These were the only available geoscience data for assessing the well position in the stratigraphy.

Descriptions of cuttings lithologies indicate a change in lithologies approximately half-way through the here investigated section. The upper part was described as reddish-brown to dark red siltstones and very fine sandstones, and dark grey to black and greenish grey shales. In contrast, the lower section has been described as reddish-brown and grey siltstones, moderate hard to hard with a blocky to sub-blocky texture and non-calcareous. Both sections clearly suggesting sedimentary rocks (Fig. 2a).

Wellsite XRF and XRD

Based on element concentrations determined from XRF, a theoretical/synthetic gamma-ray response ($GR_{(synth.)}$) was calculated for each sample using the formula after Ellis and Singer (1987):

$$GR_{(synth.)} [API'] = 16K [\%] + 4Th [PPM] + 8U [PPM]$$

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The values are used in a log against depth (Fig. 2b) to compensation in the absence of an MWD GR log. The log provided a first estimation of the likely lithology that had been drilled. The $GR_{(synth.)}$ log suggests four separate lithology intervals ranging from low, intermediate to high GR responses.

SiO_2/Al_2O_3 ratios are utilized to differentiate sandstones from argillaceous sand-/siltstones and mudrocks, with low values indicating a likely mudrock and/or argillaceous lithology, while high values imply sandstones.

The geochemical signatures have been interpreted in a chemostratigraphic context (not subject of this paper) and five chemostratigraphy zones and six subzones have been differentiated (Fig. 2b).

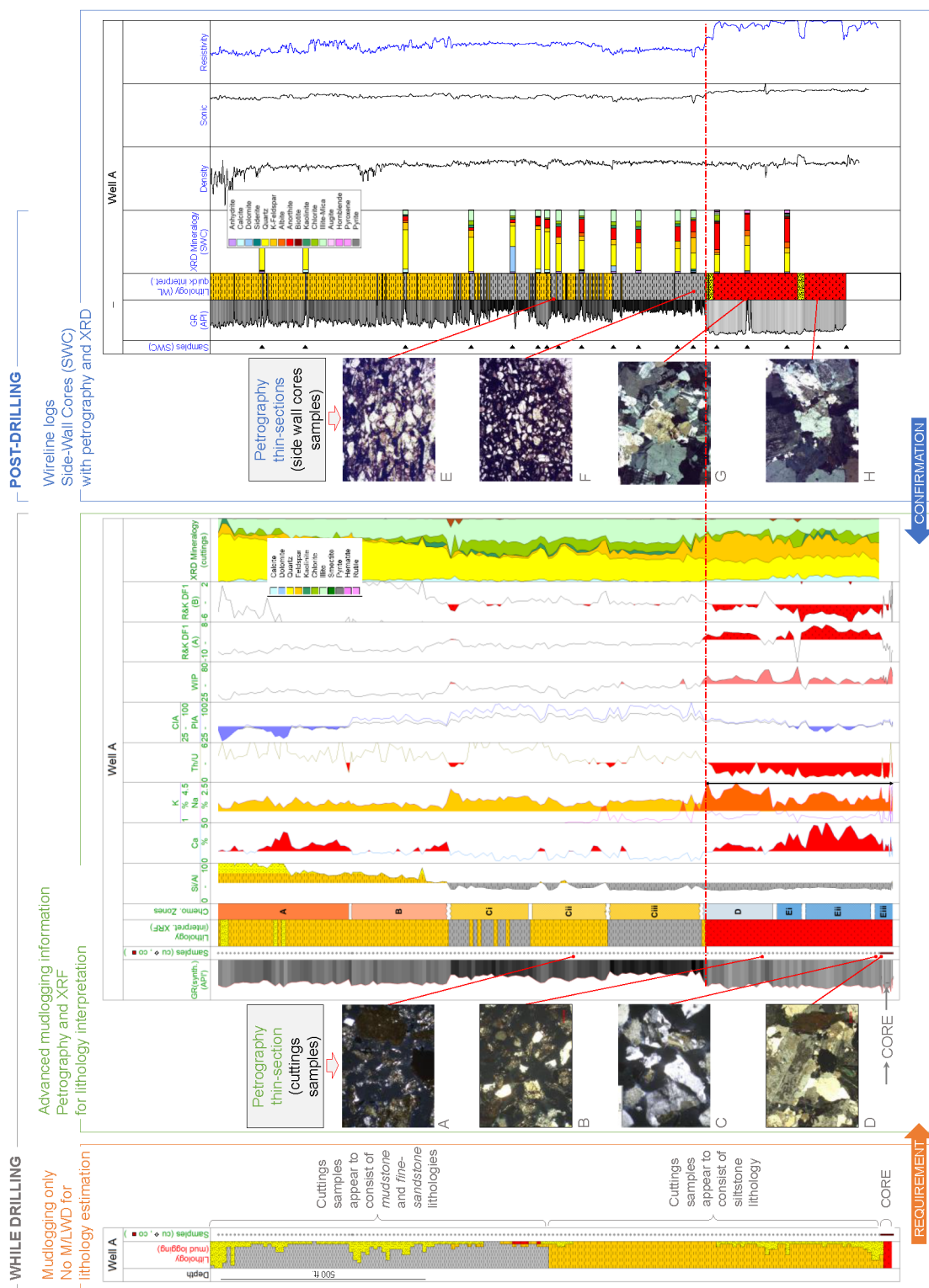


Figure 2—Summary of the stages of delineation of sedimentary and igneous rocks in the example well. Fig. 2a—Sample description only; Fig. 2b—application of wellsite XRF and XRD, and additional petrography on selected cuttings samples (while drilling) and core; Fig. 2c—detailed interpretation of lithologies, XRD and petrography on selected SWC samples, and selected wireline logs.

Igneous rocks can broadly be classified based on their mineralogical and hence geochemical compositions. A common classification is based on the silica concentrations (SiO_2) that differentiates acidic, intermediate, basic, and ultrabasic compositions with decreasing silica content (Fig.1).

Iron (FeO) and magnesia (MgO) are used as proxies for dark minerals, such as amphibole, pyroxene and olivine. Potassium (K₂O) and sodium (Na₂O) and calcium (CaO) oxides act as proxies for feldspars.

FeO + MgO values generally increase, while K₂O + Na₂O values decrease from felsic, over intermediate, mafic, to ultramafic according to the mineralogical compositions of these rock types (Fig. 1).

Amphiboles and pyroxene are usually not present in mature and deeply buried siliciclastic sediments, due to their instability towards weathering and diagenetic processes. They become dissolved and their Fe and Mg content may be incorporated into authigenic clay-minerals, such as chlorite.

The wellsite XRD instrument had a standard setup for a common mineral suite in (more mature) siliciclastic sediments. Igneous minerals such as of the amphibole and pyroxene groups, or olivine were not part of this setup.

Logs of FeO+MgO and K₂O+Na₂O, as suggested by the igneous rocks' classification summarized in Fig. 1, however, do not differentiate igneous from siliciclastic sediments. This is likely due to the abundance of chlorites hosting Fe and Mg in the sedimentary intervals, as indicated by wellsite XRD for the same samples. The sedimentary siliciclastics may have been derived from the same igneous rocks they overlay.

Monitoring the abundance of Ca, K, and Na on themselves, however, show a distinct change for the lower third of the evaluated section coinciding with a low GR_(synth.) response.

Ca concentrations increase. Samples are described as non-calcareous, though XED suggests some calcite to be present. Low Sulfur (S) concentrations disallow the presence of significant amounts of anhydrite. Thus, it is presumed that Ca is likely be hosted in Ca-plagioclases.

At the same depth, K concentrations decrease, while amounts of Na increase. This is interpreted as a decrease in K-feldspar and or illite, and an increase in Na-bearing plagioclases. The presence of Ca- and Na-plagioclases imply relatively unaltered igneous rocks.

Chemical weathering indices. Chemical weathering indices potentially show differences between unweathered igneous rocks and sedimentary rocks. The most popular indices in literature are the Chemical Index of Alteration (CIA; Nesbit and Young 1982), Plagioclase Index of Alteration (PIA; Fedo et al. 1995), and the Weathering Index of Parker (WIP; Parker 1970) (Table 1). CIA and PIA contain a CaO* component, which refers to CaO from the siliciclastic fraction only. A correction for apatite following Fedo et al. (1995) has been applied. Furthermore, based on XRD calcite values, a stoichiometric correction of CaO has been applied.

Table 1—Chemical weathering/alteration indices applied in this study. Note, all oxides are converted to molecular proportions, as stated in the original publications. CaO* denotes CaO from the siliciclastic fraction only.

Chemical Index of Alteration (CIA)	Nesbit and Young, 1982
$CIA = 100 \times (Al_2O_3 / (Al_2O_3 + CaO^* + Na_2O + K_2O))$	
Plagioclase Index of Alteration	Fedo et al., 1995
$PIA = 100 \times (Al_2O_3 - K_2O) / (Al_2O_3 + CaO^* + Na_2O - K_2O)$	
Weathering Index of Parker	Parker, 1970
$WIP = 100 \times (2 \times Na_2O / 0.35) + (MgO / 0.9) + (2 \times (K_2O / 0.25) + (CaO) / 0.7)$	

Fresh (unaltered) igneous rocks have CIA and PIA values of ≤ 50 (Nesbit and Young, 1982; Fedo et al., 1995). Fig 2b shows that such vales are reached in the upper and lower quarter of the evaluated section. It is unlikely that the upper quarter of the section consists of igneous rocks, but possibly contains a higher number of igneous clasts, which broadly coincides with higher feldspar content from XRD.

Only the WIP shows an exclusive change in values at the lower part of the investigate section coinciding with notable changes in all other geochemical logs (Fig. 2b). Although Parker (1970) states values of >100 for 'fresh' rocks, a cut-off of >50 works in this example.

Th/U ratios. An additional geochemical element ratio that indicates a distinct change for the igneous rock section is the Th/U ratio. Th/U ratios display lower values in the interval interpreted as igneous than in overlaying sedimentary rocks (Fig. 2b). The low values are due to low Th concentrations, while U values are relatively consistent through the hole section. Thus, the Th/U ratios not only differentiate the igneous from sedimentary rocks, they also suggest a more mafic rock type, as Th is more abundant in felsic rocks compared to mafic rocks.

Discriminant Functions. Discriminant Functions (DF) after Roser and Korsch (1988) have been applied to the data. The DF's are usually presented in two binary diagrams intending to differentiate quartzose provenance from felsic, intermediate, and mafic provenances.

Table 2—Discriminant Function (DF) formulas for the two DF diagrams after Roser and Korsch (1988).

Discriminant Functions for diagram A (Fig. 3a):	
DF1(A)	$-1.773x \text{ TiO}_2 + 0.607x \text{ Al}_2\text{O}_3 + 0.76x \text{ Fe}_2\text{O}_{3(\text{total})} - 1.5x \text{ MgO} + 0.616x \text{ CaO} + 0.509x \text{ Na}_2\text{O} - 1.224x \text{ K}_2\text{O} - 9.09$
DF2(A)	$0.445x \text{ TiO}_2 + 0.07x \text{ Al}_2\text{O}_3 - 0.25x \text{ Fe}_2\text{O}_{3(\text{total})} - 1.142x \text{ MgO} + 0.438x \text{ CaO} + 1.475x \text{ Na}_2\text{O} + 1.426x \text{ K}_2\text{O} - 6.861$
Discriminant Functions for diagram B (Fig. 3b):	
DF1(B)	$30.638x \text{ TiO}_2/\text{Al}_2\text{O}_3 - 12.541x \text{ Fe}_2\text{O}_{3(\text{total})}/\text{Al}_2\text{O}_3 + 7.329x \text{ MgO}/\text{Al}_2\text{O}_3 + 12.031x \text{ Na}_2\text{O}/\text{Al}_2\text{O}_3 + 35.402x \text{ K}_2\text{O}/\text{Al}_2\text{O}_3 - 6.382$
DF2(B)	$56.500x \text{ TiO}_2/\text{Al}_2\text{O}_3 - 10.879x \text{ Fe}_2\text{O}_{3(\text{total})}/\text{Al}_2\text{O}_3 + 30.875x \text{ MgO}/\text{Al}_2\text{O}_3 - 5.404x \text{ Na}_2\text{O}/\text{Al}_2\text{O}_3 + 11.112x \text{ K}_2\text{O}/\text{Al}_2\text{O}_3 - 3.89$

Samples with an igneous interpreted origin are separated from a quartzose sedimentary provenance in the diagram of the first two DF's (Fig. 3a). Furthermore, the samples are classified as of a mafic to intermediate igneous provenance. The second diagram is more ambiguous though the interpreted igneous samples mainly plot in the mafic igneous provenance field. The first set of DF's have been found more reliable in differentiating igneous from sedimentary rocks in other studies, too.

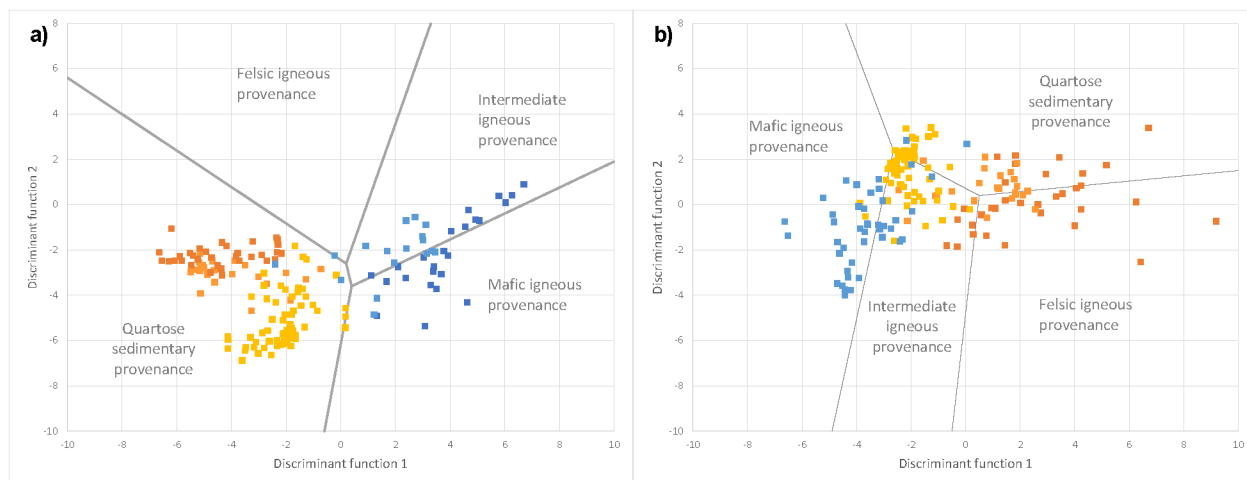


Figure 3—Discriminant Function diagram 1 after Roser and Korsch (1988). Colors of data points refer to the colors of Chemo. Zones in Fig.2 (orange-red = A, pale-red = B, orange = C, light blue = D, blue = E). Fig. 3a—DF diagram A separates zones A-C from zones D and E, and suggest an intermediate to mafic igneous source. Fig. 3b—DF diagram B separates zones A-C from D and E, but suggests intermediate and felsic provenances for the sedimentary rocks.

When displayed as logs, however, DF1 of both diagrams are of better use than DF2 of either diagram. DF1 of the first diagram (A) and of the second diagram (B) are displayed in Fig. 2b and mark the section interpreted as igneous.

TAS - Total Alkalies vs. Silica diagram for plutonic rocks. For a further determination of the likely type of igneous rock the Total Alkalies vs. Silica (TAS) diagram for plutonic rocks, after [Le Maitre et al. \(2005\)](#), has been utilized. The geochemical data from chemostratigraphy zones D and E, plotted in the TAS diagram ([Fig. 4](#)), indicate diorite to quartz-diorite (grano-diorite) as the probable plutonic rock type.

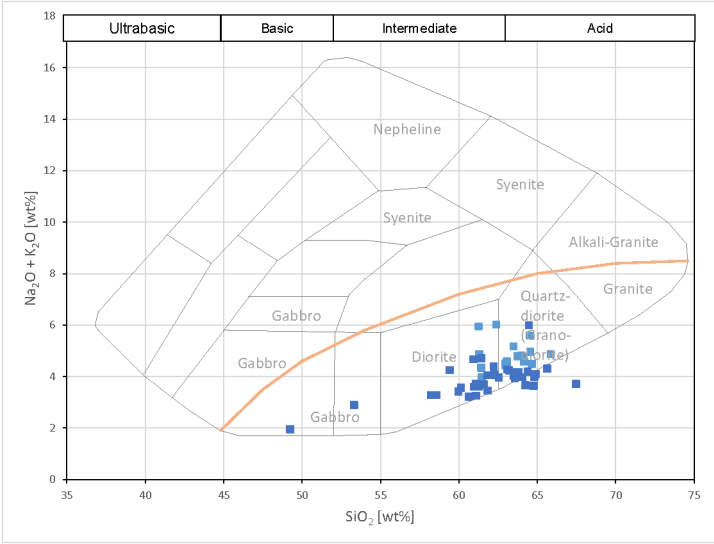


Figure 4—TAS diagram for plutonic rocks after [Le Maitre et al. \(2005\)](#). The igneous/plutonic rocks data (zones D and E) plot within the diorite and quartz-diorite fields, while two samples fall within the gabbro field. Samples plotting slightly outside the diagram are assumed to be affected by sample quality (e.g., mix with bore hole cavings).

Petrography thin section

Although rather uncommon, petrographic thin-section analyses have been applied on cuttings samples while drilling. Cuttings samples were regularly sent to the company’s core laboratories, where thin-sections for petrography were prepared. The petrography descriptions are summarized in [Table 3](#).

Table 3—Summary of petrographic descriptions of thin sections from cuttings, SWC’s and bottom core samples. Letters A to D refer to the sample positions as shown in [Fig. 2b](#) and [Fig. 2c](#), from top to bottom.

Petrography	Source
A) oxidized silty/sandy mudstone: subangular quartz and feldspars	cuttings
B) altered igneous rock: quartz, feldspar and igneous fragments	cuttings
C) questionable diorite: igneous components with interlocking quartz, feldspars and altered hornblende	cuttings
D) hornblende diorite: interlocking plagioclase, quartz, some K-feldspar, and hornblende	core
E) cross-laminated argillaceous sandstone: quartz, feldspar, mica, clay pellets, heavy minerals and lithic fragments	SWC
F) very fine to medium grained, argillaceous sandstone: quartz, feldspar, mica, lithic fragments, heavy minerals	SWC
G) diorite: quartz, K-feldspar and plagioclase, small amounts of biotite, hornblende and other	SWC
H) diorite: quartz, feldspars, ferromagnesian minerals (biotite, hornblende), sphene, and other	SWC

The first strong indication for penetration of igneous rocks from petrography on cuttings samples occurs with sample C (see [Fig. 2b](#)), which is described as a possible diorite. These descriptions of cuttings complement later petrographic analyses from SWC’s (see below).

After reaching the terminal depth of the well a bottom core was cut. A petrographic thin-section (sample D) from the core is described as a hornblende diorite.

Post drilling analyses

Several SWC's were produced after the well terminated and logged with wireline tools. Selected SWC were analyzed by XRF in the in-house core laboratories. The XRF mineralogy reveals high concentrations of anorthite in the lower third of the evaluated section confirming the initial interpretation of abundant Ca-plagioclase from XRF Ca concentrations.

Thin-section analyses classify the magmatic rocks in the SWCs from this section as diorite (Table 3).

Wireline logs provided a detailed GR trace that validates the calculated $GR_{(synth.)}$ calculated from XRF derived K, Th, and U concentrations (Fig. 2b and Fig. 2c).

Density and Sonic logs are non-diagnostic in this case for the entry into the magmatic basement rocks, while the Resistivity log shows a notable increase in values.

Conclusions

The study demonstrates that geochemical and mineralogical data generated from ditch cuttings and interpreted in near-real time, e.g. at the wellsite, can delineate sedimentary and igneous lithologies during drilling without MWD tools.

Geochemical proxies for different feldspar types (K, Na, and Ca) show significant changes in their logs that indicate the presence of igneous rocks in the lower third of the investigated section. This interpretation is congruent with an increase in feldspar content from wellsite XRD mineralogy suggesting less weathering/alteration as expected for magmatic basement rocks.

In addition, WIP values indicate less weathering of the lower third of the section suggesting fresher igneous rocks.

Furthermore, geochemical data suggest elevated concentration of Ca- and Na-plagioclases, anorthite and albite respectively, which is confirmed by post-drilling, lab-based XRD mineralogy from sidewall cores. Plagioclase feldspars, particularly Ca-rich varieties, are more abundant in intermediate and mafic igneous rocks.

Discriminant Functions, as proposed by Roser and Korsch (1988) differentiates an intermediate to mafic provenance for the igneous rock samples. In addition, the geochemical TAS diagram for plutonic rocks indicates diorite to quartz-diorite as the likely igneous rock type.

Petrography on thin section from cuttings, core and SWC describe the plutonic rocks as diorite confirming the interpretation from geochemical data.

Therefore, both data types, XRF elements and XRD minerals, either combined or on their own can be interpreted towards delineation of sedimentary and igneous lithologies in a geological context.

Acknowledgments

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Nomenclature

- BHA: Bottom Hole Assembly. The assembly of different components, e.g. drill bit and MWD/LWD tools at the bottom of the drill string.
- HSE: Health, safety, and environment policies. For hydrocarbon exploration, it is a common HSE policy to keep the workforce at the wellsite low.
- LWD: Logging while drilling. A set of bore hole tools that measure certain physical properties, such as natural gamma-ray (GR), density (RHOB), or resistivity of rock units, while the bore hole is drilled. Data are kept in memory and are evaluated when the tools are back at surface.

- MWD: Measurement while drilling. A set of bore hole tools that measure certain physical properties, such as natural gamma-ray (GR), density (RHOB), neutron porosity (PHI) or resistivity of rock units, while the bore hole is drilled. Data are transmitted for real time evaluation to the surface.
- PSE: Periodic System of the Elements.
- SWC: Side wall cores. SWCs are core sample that are taken from the side of a borehole after it has been drilled and logged.
- XRD: X-ray diffraction. XRD analyzers determine the mineralogical composition of materials, here rocks. The mineralogical output is variable depending on the rock type and lithology. Specialized knowledge is required to determine the correct and complete mineral suit of a rock sample.
- XRF: X-ray fluorescence. XRF analyzers determine the chemical composition of material, here rocks. Rock analyses commonly include a range of elements from Na (sodium, Z=11) to U (uranium, Z=92) in the Periodic System of the Elements (PSE).

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